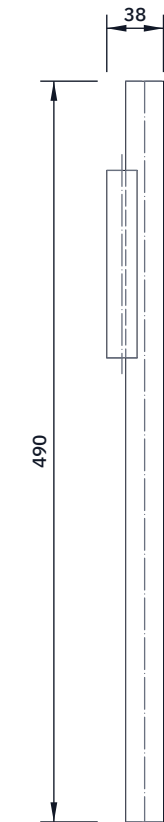


REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC
P2	Cluster shaft in 4140 quenched and tempered steel (SCM-DEC-001, item 1)	2026-10-02	AC

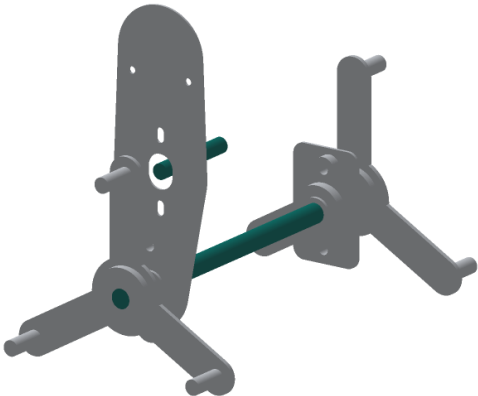


TOP VIEW
Scale 1:5; looking down (along -Z)



FRONT VIEW
Scale 1:5; looking at the front (along +Y)

RIGHT VIEW
Scale 1:5; looking at the right side (along -X)



WHERE IT GOES
This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Cluster shaft: cut 25 mm keyed shaft to 490 mm; chamfer both ends 1 mm.
4140 quenched and tempered steel, not bright 1018; keep the mill certificate.
The key runs its full length (keyway milled by the supplier).
Countershaft: cut 20 mm keyed shaft to 124 mm; chamfer both ends.
The cluster shaft runs in the two 25 mm flange bearings 336 mm apart and carries the 20-tooth sprocket inside the chain case.
Each end carries a hub: the taper-lock bush clamps it to the shaft, its outer face flush with the shaft end.
The countershaft sits 143.9 mm up the frame and 15 mm behind the cluster shaft, in two 20 mm flange bearings 86 mm apart, one on each case plate; its two sprockets sit between them.
Check: both shafts slide through their bearings by hand before the bearing grub screws are tightened.

STEPCLIMBER

github.com/BoujeeEnjinia1701/stepclimber

StepClimber cluster shaft and countershaft: making sketch

DRAWING NO. SCM-DWG-106			REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:5	UNITS mm	PROJECTION 	DATE 2026-10-02	DRAWN Amish Chadha	
MATERIAL / NOTES Cluster shaft 25 mm keyed, 4140 quenched and tempered (mill certificate); countershaft 20 mm keyed bright steel					
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